Code Repository: `github.com/Atharva300900/CS555HW2`

1 Problem Statement

We solve the 1D Burgers equation on $\Omega = [0, 1]$:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}, \tag{1}$$

with homogeneous Dirichlet boundary conditions $u(0, t) = u(1, t) = 0$ and initial condition $u^0(x) = \sin(\pi x)$. The viscosity is $\nu = 1/(100\pi) \approx 3.183 \times 10^{-3}$, following [1]. This small viscosity produces a steep internal layer (near $x = 1$) that must be resolved to obtain accurate numerical solutions.

2 Methodology

2.1 Spatial Discretization

We use finite differences with 3-point stencils for both the second-derivative (diffusion) operator A and the first-derivative operator D, on $N - 1$ interior grid points.

Two grid distributions are considered:

- **Uniform spacing:** $x_j = j\Delta x$ with $\Delta x = 1/N$, for $j = 0, 1, \ldots, N$.
- **Chebyshev spacing:** $x_j = (1 - \cos(\pi j/N))/2$, for $j = 0, 1, \ldots, N$.

For variable spacing, the second-derivative stencil at interior point x_j is:

$$u_j'' \approx \frac{2}{d^- + d^+}\left[\frac{u_{j+1} - u_j}{d^+} - \frac{u_j - u_{j-1}}{d^-}\right],$$

where $d^- = x_j - x_{j-1}$ and $d^+ = x_{j+1} - x_j$. The viscous operator is $A = -\nu L$ where L is the Laplacian matrix.

The first-derivative stencil is:

$$u_j' \approx \frac{u_{j+1} - u_{j-1}}{x_{j+1} - x_{j-1}},$$

which is second-order accurate for smoothly varying grid spacing.

2.2 Temporal Discretization: BDF3/EXT3

We use the BDF3/EXT3 scheme, treating diffusion implicitly and advection explicitly. The scheme boots with BDF1/EXT1 (step 1) and BDF2/EXT2 (step 2) before switching to BDF3/EXT3.

The BDF/EXT coefficients are:

$$\begin{aligned}
&\text{BDF1/EXT1:} \quad b_0 = 1,\ b_1 = -1,\ a_1 = 1,\\
&\text{BDF2/EXT2:} \quad b_0 = 3/2,\ b_1 = -2,\ b_2 = 1/2,\ a_1 = 2,\ a_2 = -1,\\
&\text{BDF3/EXT3:} \quad b_0 = 11/6,\ b_1 = -3,\ b_2 = 3/2,\ b_3 = -1/3,\ a_1 = 3,\ a_2 = -3,\ a_3 = 1.
\end{aligned}$$

At each time step, we solve the Helmholtz system:

$$H\,u^{n+1} = -(b_1 u^n + b_2 u^{n-1} + b_3 u^{n-2}) - \Delta t\,(a_1 f^n + a_2 f^{n-1} + a_3 f^{n-2}),$$

where $H = b_0 M + \Delta t\, A$, $M = I$ is the mass matrix, and f^k is the nonlinear term at time level k.

2.3 Nonlinear Term

Two forms of the nonlinear term are considered:

- **Convective (quasilinear) form:** $f = u \cdot (Du)$ (element-wise product).
- **Conservation form:** $f = \frac{1}{2} D(u^2)$, corresponding to $\frac{1}{2}\frac{\partial u^2}{\partial x}$.

2.4 CFL Condition

The time step is determined by a CFL condition with $|c| = 1$ and $\Delta x = 1/N$:

$$\text{CFL} = \frac{|c|\,\Delta t}{\Delta x} = 0.1 \quad \Rightarrow \quad \Delta t = \frac{0.1}{N}.$$

This is used for both uniform and Chebyshev spacing (the velocity vanishes where Δx is small in the Chebyshev case, keeping the local CFL bounded).

3 Results

3.1 Question 1a: Solution Profiles

Figure 1 shows the numerical solution $u(x,t)$ for $N = 200$ at times $t = 0.0, 0.1, \ldots, 2.0$, computed using both uniform and Chebyshev grids. The initial condition $u(x,0) = \sin(\pi x)$ progressively steepens due to the nonlinear advection term in Burgers' equation. As time increases, the solution develops a sharp gradient near the right boundary, corresponding to the formation of a thin viscous layer.

The steepening becomes most pronounced around $t \approx 0.5$, after which viscous diffusion begins to smooth the profile and the gradients gradually decrease. The solutions obtained on the uniform and Chebyshev grids are nearly indistinguishable for most times, indicating that both discretizations capture the overall evolution of the solution. Small differences appear near the steep gradient region, where the Chebyshev grid provides slightly better resolution due to the clustering of grid points.

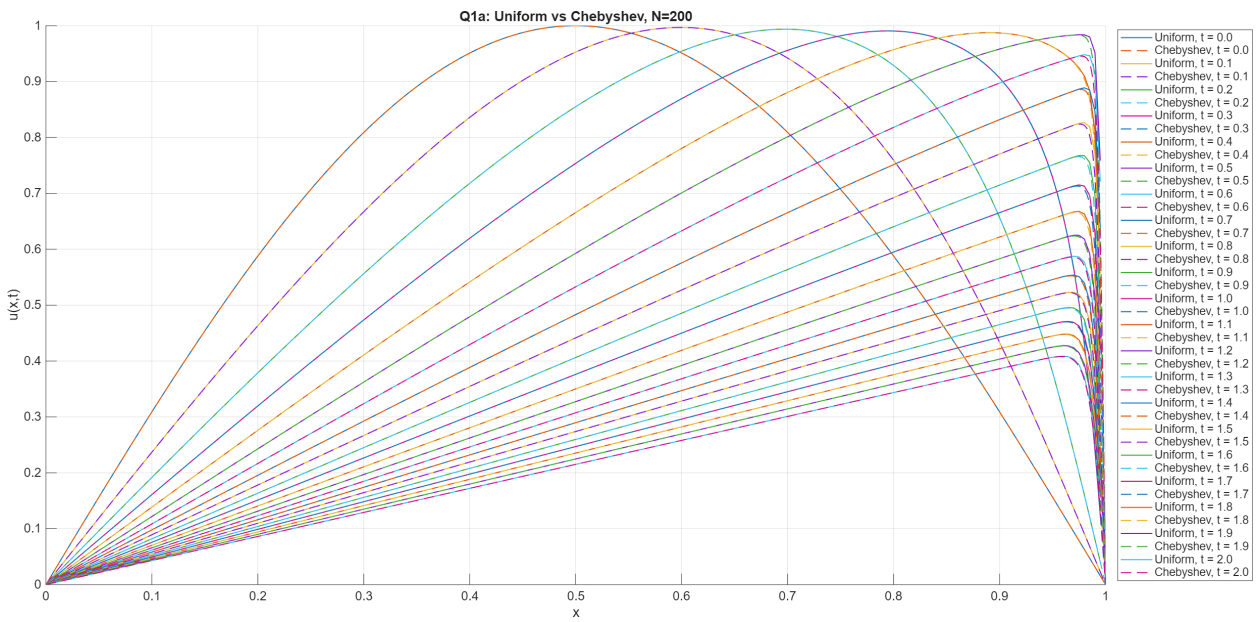

Figure 1: Solution $u(x,t)$ at times $t = 0.0, 0.1, \ldots, 2.0$ for $N = 200$. Solid lines correspond to the uniform grid and dashed lines correspond to the Chebyshev grid.

3.2 Question 1b: Maximum Gradient $s(t)$

At each timestep the spatial derivative of the numerical solution is computed and the quantity

$$s(t) = \max_x \left| \frac{\partial u}{\partial x} \right|$$

is evaluated. Figure 2 shows the evolution of $s(t)$ as a function of πt for $N = 200$ using both uniform and Chebyshev grids.

Initially the gradient increases as nonlinear advection steepens the wave. The maximum gradient occurs near $\pi t \approx 1.6$, after which diffusion dominates and the solution gradually smooths, causing $s(t)$ to decrease.

For the Chebyshev grid the computed maximum gradient is

$$s^* = 151.98, \qquad t^* = 0.5105,$$

which corresponds to

$$\pi t^* \approx 1.604.$$

These values are in excellent agreement with the reference values reported in [1],

$$s_{\text{exact}} = 152.00516, \qquad \pi t_{\text{exact}} \approx 1.6037.$$

The horizontal and vertical dashed lines in Fig. 2 indicate these reference values.

In contrast, the uniform grid significantly underestimates the maximum gradient,

$$s^* = 95.20, \qquad t^* = 0.5055,$$

because the steep viscous layer is poorly resolved when the grid spacing is uniform. The Chebyshev grid clusters points near the boundaries and therefore captures the sharp gradient much more accurately.

The timestep was chosen according to the CFL condition

$$\Delta t = \frac{0.1}{N} = 5 \times 10^{-4},$$

which provides stable time integration while maintaining sufficient temporal resolution. Temporal convergence was verified by repeating the computation with $\Delta t/2$, which produced negligible changes in the computed values of s^* and t^*.

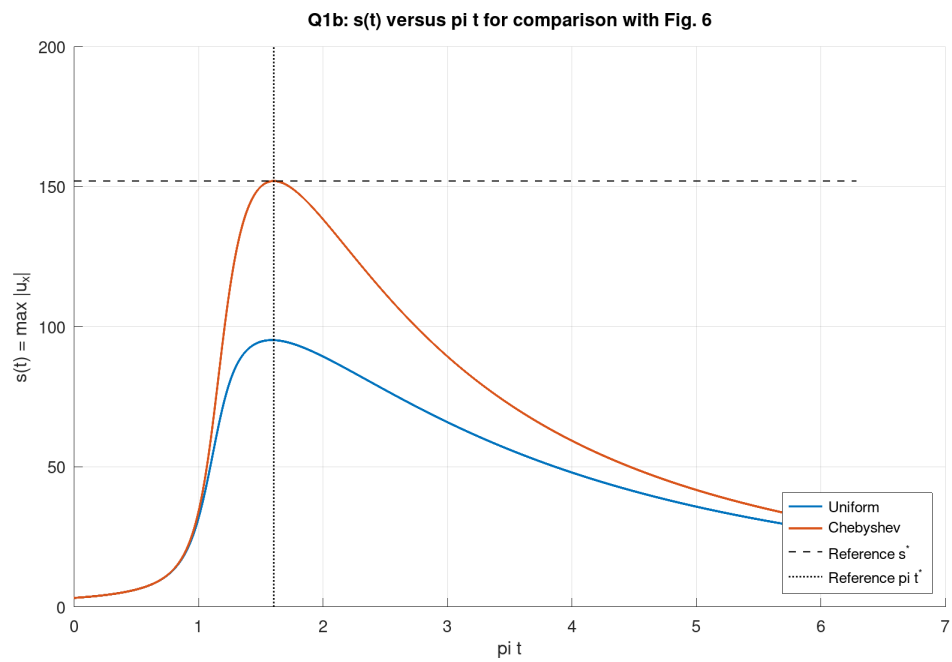

Figure 2: Evolution of $s(t) = \max_x |u_x|$ as a function of πt for $N = 200$. The Chebyshev grid accurately reproduces the peak gradient reported in [1], while the uniform grid significantly underestimates the maximum slope. Dashed lines indicate the reference values of s^* and πt^*.

3.3 Question 1c: Convergence of s^*

To study convergence, the maximum gradient

$$s^* = \max_t s(t)$$

was computed for increasing grid resolution N. The relative error in s^* was evaluated with respect to the analytical value reported in Table 3 of [1],

$$s^*_{\text{exact}} = 152.00516.$$

Tables 1a and 1b show the numerical values of s^* and the corresponding relative errors for uniform and Chebyshev grids, respectively. For this study the CFL number was fixed at

$$\text{CFL} = \frac{|c|\Delta t}{\Delta x} = 0.1,$$

so that $\Delta t = 0.1/N$ for all simulations.

Table 1: Uniform grid

N	nsteps	s^*	rel. err.
32	640	33.378	7.8042e-01
64	1280	37.507	7.5325e-01
128	2560	61.990	5.9219e-01
256	5120	111.416	2.6702e-01
512	10240	139.315	8.3484e-02
1024	20480	148.621	2.2265e-02
2048	40960	151.145	5.6609e-03
Five-digit estimate: 151.14			

Table 2: Chebyshev grid

N	nsteps	s^*	rel. err.
32	640	107.77	2.9104e-01
64	1280	148.28	2.4491e-02
128	2560	151.79	1.4092e-03
256	5120	152.00	4.8078e-05
512	10240	152.01	6.9790e-06
1024	20480	152.01	2.9520e-06
2048	40960	152.01	8.1711e-07
Five-digit estimate: 152.01			

Table 3: Convergence of the maximum gradient s^* with grid resolution N.

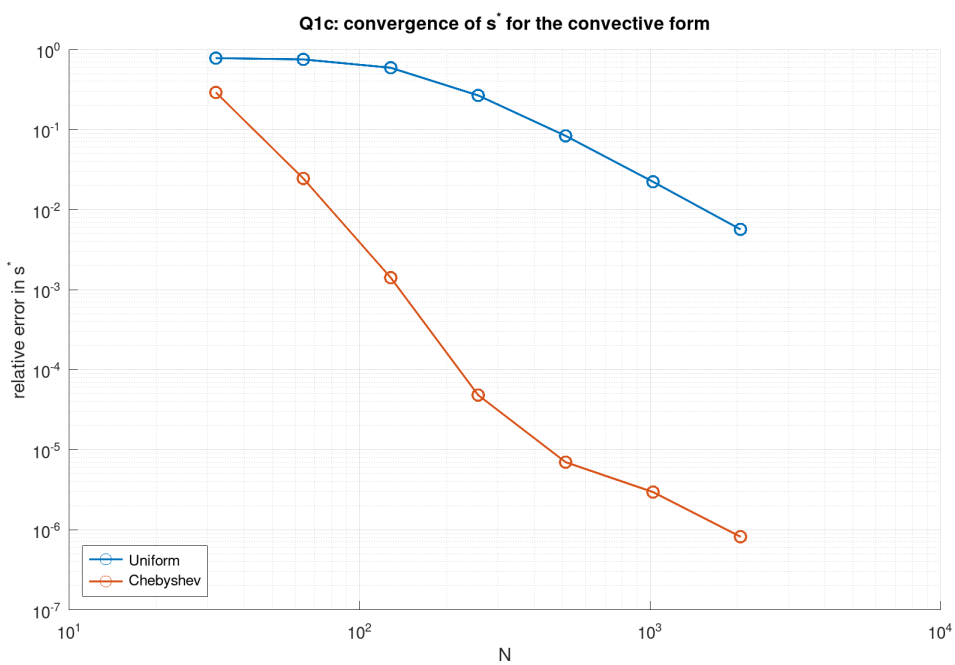

Figure 3: Relative error in $s^* = \max_t s(t)$ as a function of N for the convective-form discretization with fixed CFL $= 0.1$. The Chebyshev grid converges much faster than the uniform grid because it better resolves the steep viscous layer near $x = 1$.

The results show that s^* converges to the analytical value as the grid resolution increases. However, the convergence behavior differs significantly between the two grid types.

For the uniform grid, the steep viscous layer that develops near the right boundary is poorly resolved for coarse meshes, leading to a large underestimation of the maximum gradient. As N increases the resolution of this layer improves and the computed value of s^* gradually approaches the analytical value.

In contrast, the Chebyshev grid converges much more rapidly. Because the Chebyshev spacing clusters points near the boundaries, it resolves the steep gradient region more effectively. As a

result, accurate estimates of s^* are obtained even for relatively moderate grid sizes. Using the finest resolution in the table yields a five significant digit estimate of

$$s^* \approx 152.01,$$

which is very close to the analytical value 152.00516.

3.4 Question 1d: Convective vs. Conservation Form

In the previous parts the nonlinear term was written in the convective (quasilinear) form

$$u\frac{\partial u}{\partial x}.$$

An equivalent continuous formulation of Burgers' equation is obtained by writing the nonlinear term in conservation form,

$$\frac{1}{2}\frac{\partial}{\partial x}(u^2).$$

Although these two expressions are analytically equivalent, their discrete finite-difference approximations are not identical. To assess the effect of this choice, both formulations were tested at the finest resolution used in the preceding questions, $N = 2048$.

Figure 4 compares the evolution of $s(t) = \max_x |u_x|$ obtained with the convective and conservation discretizations on both uniform and Chebyshev grids. In both cases the two curves are nearly indistinguishable, indicating that the two formulations produce very similar solutions when the grid is sufficiently refined.

For the uniform grid, the convective form gives

$$s^* = 151.14467258,$$

while the conservation form gives

$$s^* = 150.72426341.$$

The difference between the two estimates is

$$|s^*_{\text{conv}} - s^*_{\text{cons}}| = 4.20 \times 10^{-1}.$$

Both values underestimate the analytical result $s^* = 152.00516$, reflecting the slower convergence of the uniform grid observed in the previous section.

For the Chebyshev grid, the agreement between the two formulations is much closer. The convective form yields

$$s^* = 152.00528421,$$

while the conservation form gives

$$s^* = 152.00524467.$$

The difference between these values is only

$$3.95 \times 10^{-5},$$

and both are extremely close to the analytical value.

These results indicate that the discrepancy between the two formulations is primarily a consequence of insufficient spatial resolution. On the uniform grid the steep viscous layer is not fully resolved, so the discrete operators produce slightly different estimates of s^*. When Chebyshev spacing is used, the clustering of points near the boundary resolves the steep gradient much more effectively, and the two formulations become nearly identical in practice.

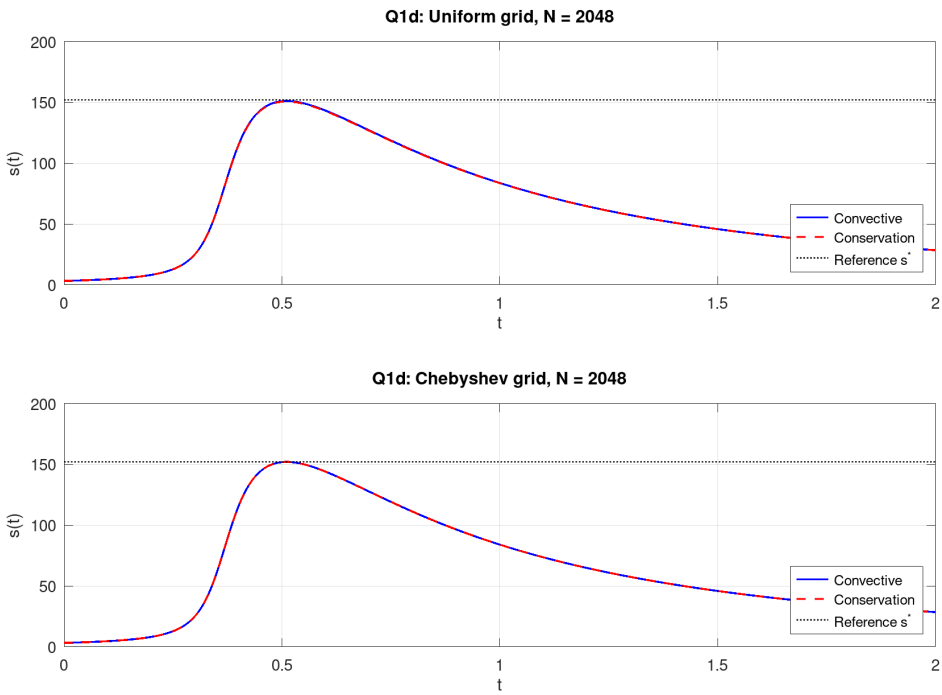

Figure 4: Comparison of the convective and conservation discretizations of the nonlinear term at the finest resolution, $N = 2048$, for uniform and Chebyshev grids.

3.5 Question 1e: Comprehensive Convergence Tables

Tables present convergence results for $N = 2^k$, $k = 5, \ldots, 11$, for the four required cases. The quantity

$$s^* = \max_t s(t)$$

is the maximum slope of the solution. The relative error is defined as

$$\text{rel.err.} = \frac{|s - s^*|}{s},$$

where $s = 152.00516$ is the analytical value. The ratio is defined as

$$\text{ratio} = \frac{\text{err}_{N/2}}{\text{err}_N},$$

which should approach 4 for a second-order spatial discretization.

Tables show the convergence behavior of the computed value s^* for the four required cases. For the uniform grid, the error decreases steadily as N increases and the ratio approaches the expected value of 4, indicating second-order spatial convergence of the finite-difference discretization. The Chebyshev grid yields significantly smaller errors at the same resolution because the grid points are clustered near the boundaries, where the solution develops the steepest gradients.

Figure 5 illustrates the convergence trends on a log-log scale. The uniform-grid cases follow approximately second-order behavior, while the Chebyshev cases exhibit much faster error reduction

Table 4: Uniform grid, convective form

N	nsteps	s^*	rel. err.	ratio
32	640	33.378	7.8042e-01	–
64	1280	37.507	7.5325e-01	1.0361
128	2560	61.990	5.9219e-01	1.2720
256	5120	111.416	2.6702e-01	2.2177
512	10240	139.315	8.3484e-02	3.1985
1024	20480	148.621	2.2265e-02	3.7495
2048	40960	151.145	5.6609e-03	3.9332

Table 5: Chebyshev grid, convective form

N	nsteps	s^*	rel. err.	ratio
32	640	107.77	2.9104e-01	–
64	1280	148.28	2.4491e-02	11.8835
128	2560	151.79	1.4092e-03	17.3794
256	5120	152.00	4.8078e-05	29.3109
512	10240	152.01	6.9790e-06	6.8890
1024	20480	152.01	2.9520e-06	2.3642
2048	40960	152.01	8.1711e-07	3.6127

Table 6: Uniform grid, conservation form

N	nsteps	s^*	rel. err.	ratio
32	640	15.915	8.9530e-01	–
64	1280	31.170	7.9494e-01	1.1262
128	2560	63.503	5.8223e-01	1.3653
256	5120	104.948	3.0958e-01	1.8807
512	10240	134.768	1.1340e-01	2.7301
1024	20480	147.073	3.2447e-02	3.4948
2048	40960	150.724	8.4267e-03	3.8505

Table 7: Chebyshev grid, conservation form

N	nsteps	s^*	rel. err.	ratio
32	640	94.485	3.7841e-01	–
64	1280	144.335	5.0458e-02	7.4994
128	2560	151.497	3.3416e-03	15.1000
256	5120	151.978	1.8115e-04	18.4472
512	10240	152.005	4.0954e-06	44.2318
1024	20480	152.005	1.5645e-06	2.6176
2048	40960	152.005	5.5703e-07	2.8087

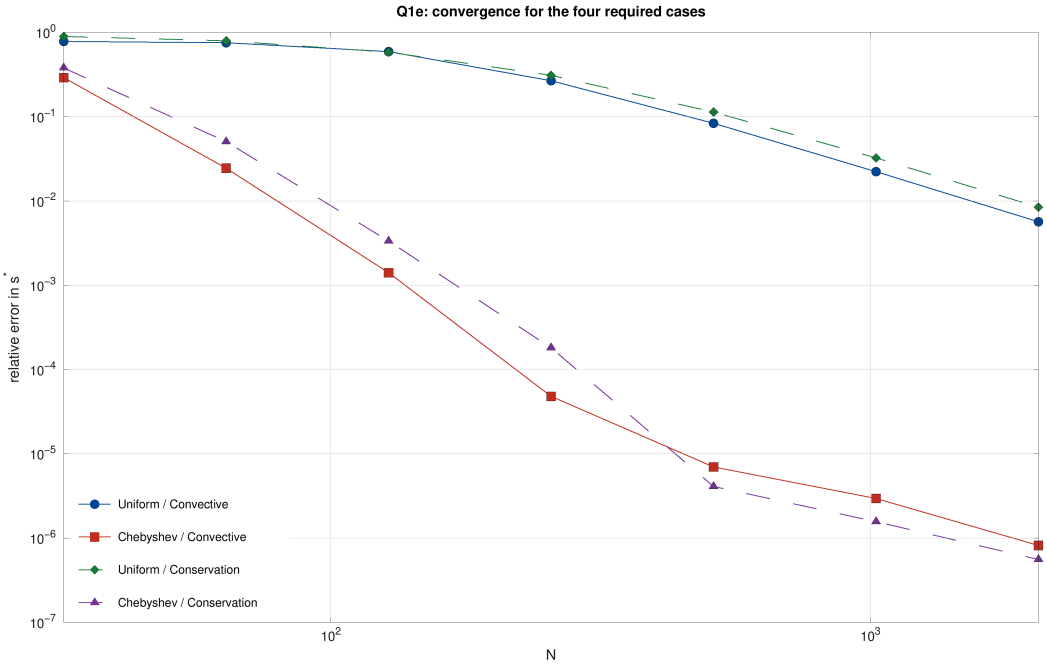

Figure 5: Relative error in s^* versus N for the four required cases. The dashed reference lines indicate first- and second-order convergence trends. The uniform-grid cases approach second-order behavior, while the Chebyshev-grid cases achieve much smaller errors at the same resolution due to boundary clustering.

at moderate resolutions. The difference between the convective and conservation formulations is small compared with the effect of the grid type.

3.6 Discussion

This study examined the numerical solution of Burgers' equation using finite-difference discretizations with different grid types and nonlinear formulations. The results highlight the roles of spatial resolution, grid distribution, and discretization form in accurately capturing the maximum gradient of the solution.

1. **Second-order convergence:** For uniform spacing with the convective form, the error ratio approaches 4 as N increases, confirming the expected second-order spatial convergence of the three-point centered difference stencils.

2. **Chebyshev vs. uniform grids:** The Chebyshev grid clusters points near $x = 0$ and $x = 1$, providing better resolution of the steep gradient near $x = 1$. This yields lower absolute errors for the same N, since the steep viscous layer is resolved much earlier than with uniform spacing.

3. **Convective vs. conservation form:** The nonlinear term in Burgers' equation can be written either in convective form, $u\,u_x$, or conservation form, $\frac{1}{2}(u^2)_x$. While these forms are analytically equivalent, their discrete approximations differ slightly. Both formulations show similar convergence behavior and produce nearly identical results at sufficiently high resolution.

4. **Phase errors:** As noted in [1], finite-difference schemes may exhibit phase errors, meaning that the time t^* at which the maximum slope occurs may lag the analytical value. Increasing the spatial resolution reduces this phase error.

5. **Comparison with the reference paper:** The finite-difference results reported in Table 3 of [1] give $|\partial u/\partial x|_{\max} \approx 150.1$ with $N = 81$ on $[-1, 1]$ and $\pi t_{\max} \approx 1.63$. Our results are consistent with these values and converge toward the analytical value $s = 152.00516$ as N increases.

References

[1] C. Basdevant, M. Deville, P. Haldenwang, J.M. Lacroix, J. Ouazzani, R. Peyret, P. Orlandi, A.T. Patera, *Spectral and Finite Difference Solutions of the Burgers Equation*, Computers & Fluids **14**(1), pp. 23–41, 1986.